An Integrated multi-scale model for pediatric brain tumor survival prediction

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Motivation

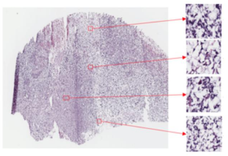
Brain tumors are the most common solid tumors affecting children, and its prognosis has been a great challenge for physicians and researchers. With the advances in high-throughput sequencing technology and digital pathology, more quantitative data is now becoming available and more information may potentially be discovered in whole slide images (WSIs) and molecular tumor characteristics to determine survival and treatment. Imaging and genomic data, though very different in nature, both may contain different aspects of disease characteristics that are important for survival prediction. Hence our work aims to build a framework to integrate two data modules, whole-slide histopathology image data, and RNA sequencing data, for a unified model to improve pediatric brain tumor survival outcome prediction.

Fundamentals

The target of this study is to predict survival outcome for pediatric brain tumor patients with two data modules:

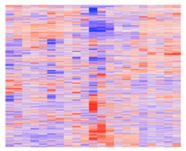
- Histopathology data – Whole slides image
- Gene expression data - RNA sequencing

• Histopathology data



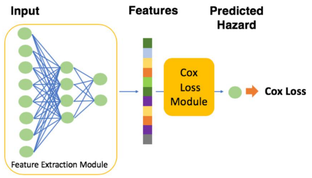
198 samples, 442 whole slide images

• Gene expression data



198 samples, 12778 genes

The imaging data and genomic data are both of high dimensions. The survival prediction framework is based on a network consisting of two modules: the feature extraction module and the cox loss module. The survival outcome is predicted by the cox loss module, which takes as input the feature outputs from the feature extraction module.



The Cox Loss, traditionally used in survival regression, is a differentiable loss that can be used to train deep neural networks. The parameters of the Cox loss module "beta" are optimized by minimizing the negative of the partial log-likelihood:

$$\text{loss} = -\log L(\beta) = -\sum_{Y_i, uncensored} [\beta - \log(\sum_{Y_j \geq Y_i} e^{\beta \cdot X_j})]$$

Data fusion survival prediction

Evaluation methods

We evaluate the survival prediction model performances with one commonly used evaluation metrics: the concordance index (C-index. The C-index is a standard performance measure of a model's predictive ability in survival analysis. It is calculated by dividing the number of all pairs of subjects whose predicted survival times are correctly ordered, by the number of pairs of subjects that can be ordered. A pair cannot be ordered if the earlier time in the pair is censored or both events in the pair are censored. A value of 1.0 indicates perfect prediction where all the pairs are correctly ordered, and a value of 0.5 indicates random prediction.

Methods

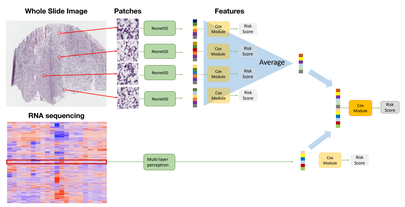
Feature extraction for histopathology and gene expression data

We use a 50 layer Resnet pretrained on the ImageNet to extract features from the histopathology data, and the resulting feature vectors are of size 2048 for each patch. The feature vector for a Whole-Slide Image is the average feature vector of the patches. For the genomic data, we use a multi-layer perceptron with ReLu activation functions and 1 hidden layer to extract features. This yields a 2048-dimensional genomic feature vector per patient.

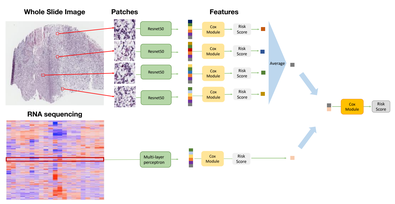
Multi-scale fusion model

We propose three data fusion strategies to integrate histopathology image data and genomic data: a feature fusion, a late fusion and a joint training fusion. The first strategy is called feature fusion, whereby we capture the respective survival features extracted from the imaging and genomic data, i.e. the inputs to the last fully connected layers in each of the survival prediction models. We concatenate the extracted features and then train a third survival prediction MLP using the merged features as input. The second strategy is late fusion which implements fusion at the output stage. We combine the predicted risk scores from each model, i.e. the outputs of the last fully connected layers in each model, and then train a final cox regression model using the merged scores. The third strategy is joint training fusion. We simultaneously train the histopathology and gene expression data modalities, and obtain a joint feature that is fed into a fully connected layer implementing the Cox loss.

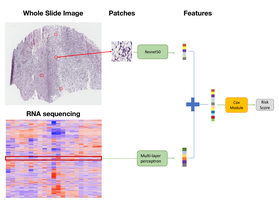
Feature fusion



Late fusion



Joint training fusion



Results

Data fusion survival prediction

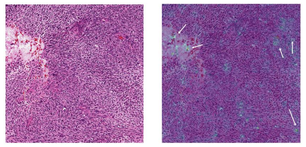
We observe improved predictive performance for overall survival prediction from data fusion methods on both adult and pediatric multi-scale data cohorts, demonstrating that integration may have discovered additional information that is not revealed in either modality alone.

Cohort	Data	Method	Concordance Index	Integrated Brier Score
Adult glioma	Pathology data	Single modality	0.75	0.16
	RNA data	Single modality	0.78	0.15
	Clinical data (age, sex)	Single modality	0.77	0.14
	Pathology + RNA	Feature fusion	**0.79**	**0.13**
	Pathology + RNA	Late fusion	**0.79**	0.14
	Pathology + RNA	Joint training fusion	**0.79**	0.17
Pediatric glioma	Pathology data	Single modality	0.79	0.14
	RNA data	Single modality	0.91	**0.08**
	Clinical data (age, sex)	Single modality	0.54	0.22
	Pathology + RNA	Feature fusion	0.90	0.09
	Pathology + RNA	Late fusion	0.91	**0.08**
	Pathology + RNA	Joint training fusion	**0.95**	**0.08**

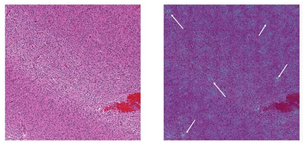
Feature importance

We further investigate how the models can be interpreted by examining the relative importance of features with respect to the prediction. As one example, we use the joint training fusion model on pediatric data, which achieves the best result among competitors. We backpropagate the predicted risk and compute the gradient of each input feature. The joint training model takes the histopathology images and the gene expression vectors as input simultaneously. For the images, we visualize the important pixels on two example images, one with good survival, and one with poor survival, by plotting saliency maps that highlight the locations with larger absolute gradient values, which signifies greater influence on how the prediction is made

Poor survival



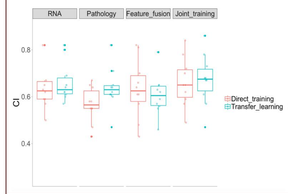
Good survival



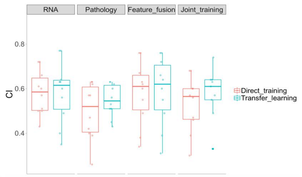
Transfer learning

we investigate transfer learning from glioma to medulloblastoma and ependymoma. medulloblastoma and ependymoma are less common subtypes of pediatric brain tumor, and hence the number of available samples for each subtype is much smaller compared to glioma. We study whether it is possible to use transfer learning to improve prediction results when the training sample is limited. We compare the results of direct training, where models are trained directly with medulloblastoma or ependymoma training dataset, and transfer learning, where models are pre-trained with glioma data, and then finetuned with medulloblastoma or ependymoma training dataset.

Medulloblastoma



Ependymoma



Conclusions

We have proposed a framework to integrate two types of biomedical data: whole slide histopathology images and RNA sequencing data, to build a unified prediction model for pediatric brain tumor survival analysis. We build independent models with different data modules, and use both separate model fusion and joint fusion. We demonstrate that our method enables the integrated model to achieve synergistic performances and reach better prediction results than using each single module alone. We find that the two fusion methods produce similar results for the integrated analysis, and further work is needed to thoroughly analyze the effects of different approaches to model fusion. In future works we plan to extend this analysis to add other modules to capture additional data types including radiology images and microRNA sequencing data, to evaluate whether these additional data modalities can improve the predictive performance.

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